2022

COMPUTER SCIENCE — GENERAL

Paper : SEC-A-1

(Communication, Computer Network and Internet)

Full Marks : 80

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question nos.*** **1** and **2** and ***any four*** questions from the rest.

1. Answer ***any ten*** questions : 2×10

(a) What is Baud rate?

(b) What are the functions of a router?

(c) Why is default gateway used?

(d) How are TCP and UDP different?

(e) What is channel capacity?

(f) What is Geostationary satellite?

(g) Mention two advantages of Star topology.

(h) What are the drawbacks of client-server model?

(i) State the range of Class B IP address.

(j) What is MAC address?

(k) What are the different layers of TCP/IP model?

(l) What is attenuation?

(m) What is URL?

(n) What is NAT?

(o) Mention few features of Delta modulation.

2. Write short notes on (***any four***) : 5×4

(a) VSAT

(b) QAM

(c) ISDN

Please Turn Over

(d) OSI model

(e) FTP

(f) Amplitude Modulation.

3. (a) Briefly describe how FDM works.

(b) Assume that a voice channel occupies a bandwidth of 4 kHz. We need to multiplex 10 voice channels with guard bands of 500 Hz using FDM. Calculate the required bandwidth.

(c) What is frequency shift keying? 5+3+2

4. (a) State the differences between connectionless and connection oriented services.

(b) Discuss different characteristics of WAN.

(c) Explain Nyquist theorem. 4+3+3

5. (a) Encode the string 01001110 in NRZ-L and Differential Manchester encoding scheme. Clearly mention the conventions.

(b) Compare between ASK and FSK.

(c) What is bit length? 4+4+2

6. (a) What is SNR? Define its unit.

(b) An analog signal carries 4 bits/signal element. If 1000 signal elements are sent per second, find the bit rate.

(c) Mention advantages of optical fibre for data communication.

(d) State the relationship between SNR and channel capacity of a noisy channel. (2+1)+2+3+2

7. (a) How are Public IP different from Private IP?

(b) What are the basic functions of DNS server?

(c) Briefly discuss the structure of a data packet. 3+2+5

8. (a) What are the differences between IPV4 and IPV6 addressing?

(b) How does the web browser reach an IP address? Use schematic diagram to explain the process. 5+5

9. (a) What is search engine?

(b) What is the purpose of multiplexing?

(c) Briefly explain different services provided by the Application layer in internet model. 2+4+4